Portfolio Optimization & Quantitative Trading with LLMs

Lecture 10 — Signals, Backtests, and Risk

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Outline

The Central Question — and a Stubborn Paradox

THE BIGGER PICTURE

Given N risky assets, how should a rational investor allocate wealth? Markowitz (1952) reduced this to a quadratic program: the *efficient frontier*. Decades of machinery followed — yet a naive $1/N$ equal-weight portfolio **routinely beats** the "optimal" one (DeMiguel et al., 2009).

Why does more sophistication underperform?

- The theory is right; the *inputs* are noisy.
- Estimating $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ from data dominates the gains.
- This is the opening for LLMs: extract *more precise* $\boldsymbol{\mu}$ from text.

Thesis of the lecture

LLMs are **information-extraction engines**, not optimizers. They produce signals; classical theory dictates how those signals become portfolio weights.

Markowitz Mean-Variance Optimization

UNDER THE HOOD

Let $\mathbf{r} \in \mathbb{R}^N$ be excess returns, $\boldsymbol{\mu} = \mathbb{E}[\mathbf{r}]$, $\boldsymbol{\Sigma} = \mathrm{Cov}[\mathbf{r}]$ (positive definite). For weights $\mathbf{w}$:

$$\mu_p = \boldsymbol{\mu}^\top \mathbf{w}, \qquad \sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w}.$$

Efficient portfolio at target return $\bar{\mu}$:

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w} \quad \text{s.t.} \quad \boldsymbol{\mu}^\top \mathbf{w} = \bar{\mu}, \ \ \mathbf{1}^\top \mathbf{w} = 1, \ \ \mathbf{w} \geq \mathbf{0}.$$

Maximum-Sharpe (tangency) portfolio, unconstrained closed form:

$$\mathbf{w}^{\mathrm{SR}} \ \propto \ \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}.$$

Scales *inversely* with risk, *linearly* with the return signal.

Why Estimation Error Wins

The inputs are the problem:

- Sample $\hat{\boldsymbol{\mu}}$ is noisy: with 60 months of data the SE of a monthly mean is $\approx \hat{\sigma}/\sqrt{60}$ — on the order of the mean itself.
- Sample $\hat{\boldsymbol{\Sigma}}$ is near-singular when N is large vs. T; its inverse *amplifies* errors.
- Result: heavy bets on recently lucky assets $\rightarrow$ unstable, poor out-of-sample.

Standard remedies:

- *Shrinkage* — pull $\hat{\boldsymbol{\Sigma}}$ toward a structured target.
- *Robust optimization* — replace point estimates with uncertainty sets.
- *Factor models*: $\boldsymbol{\Sigma} = \mathbf{BFB}^\top + \mathbf{D}$, reducing $N(N+1)/2$ entries to K factors (Fama–French 3- / 5-factor).

The DeMiguel et al. (2009) finding

Across **14 datasets**, $1/N$ beats sophisticated optimizers on out-of-sample Sharpe. Precise $\boldsymbol{\mu}$ is the missing ingredient — enter LLMs.

Black–Litterman: Equilibrium Prior + Views

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Don't trust raw sample means. Start from a prior that reflects *market equilibrium*, then update it with investor views via Bayes. This decouples *where* you have a view from *what* the view is — exactly the shape an LLM signal takes.

UNDER THE HOOD

Equilibrium implied returns from market-cap weights $\mathbf{w}^{\text{mkt}}$, risk aversion δ:

$$\boldsymbol{\Pi} = \delta\,\boldsymbol{\Sigma}\,\mathbf{w}^{\text{mkt}}, \qquad \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\Pi}, \tau\boldsymbol{\Sigma}).$$

K views $\mathbf{P}\boldsymbol{\mu} = \mathbf{q} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$. Posterior mean:

$$\hat{\boldsymbol{\mu}} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1}\left[(\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\Pi} + \mathbf{P}^\top\boldsymbol{\Omega}^{-1}\mathbf{q}\right].$$

No views $\Rightarrow \hat{\boldsymbol{\mu}} \to \boldsymbol{\Pi}$, optimum reduces to the market portfolio (sanity check).

Black–Litterman: A Single LLM View

Example — Apple absolute view

An LLM reads Apple's earnings call $\rightarrow$ 0.72 probability of an EPS beat. Translate to a BL view:

$$\mathbf{P} = \mathbf{e}_{\mathrm{AAPL}}^{\top}, \quad q = 1\% \text{ (monthly excess)}, \quad \Omega_{11} = (0.5\%)^2.$$

Prior $\Pi_{\mathrm{AAPL}} = 0.3\%$. The posterior shifts upward by an amount $\propto (q - \mathbf{P\Pi})$ relative to total uncertainty $\tau \mathbf{P \Sigma P}^{\top} + \mathbf{\Omega}$.

- High confidence (Ω_{11} small) $\Rightarrow$ posterior $\approx q = 1\%$.
- Low confidence $\Rightarrow$ posterior barely leaves Π_{AAPL}.
- BL prevents *any single noisy signal* from dominating the portfolio.

Empirical validation: Mantshimuli & Mwamba (2025) aggregate multi-LLM sentiment views via an LSTM into BL — improving Sharpe and annualised return over single-LLM and market-cap benchmarks. Yuksel (2025) treats the LLM as an evolutionary portfolio *selector*.

Where LLMs Can — and Cannot — Add Value

Clear value

- Structuring *alternative data* (calls, filings) into factors/views at scale
- *Risk monitoring* — flagging emerging risks before they hit the statements
- *Analyst-view aggregation* with calibrated uncertainty

No clear value

- Return prediction beyond a few days (factor risk dominates)
- Replacing factor models (not built to estimate $\boldsymbol{\Sigma}$)
- Timing in isolation (no order-flow / microstructure access)

Two hypotheses underpin the case:

- *Information*: text holds un-impounded return information (Grossman–Stiglitz: efficiency cannot be perfect). Evidence *mixed* — Lopez-Lira & Tang (2023) find significant but small, costly-to-harvest signal.
- *Extraction*: LLMs read negation, hedging, sarcasm better than bag-of-words — this conjunct is largely *unambiguous* (FinBERT).

Text: The Least-Saturated Alternative Data

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Satellite imagery and card flows erode fast once adopted. **Text** — earnings calls, analyst reports, filings, news, central-bank speak — generates terabytes of natural language a year and stays comparatively un-mined. LLMs convert this prose into numerical signals at scale.

Two construction pathways in this section:

1. **Views** for Black–Litterman — a per-firm q_i with calibrated uncertainty.
2. **Factors** for Fama–French — a long-short "language factor" tested for alpha.

Extracting Analyst Views from Text

Four-stage extraction pipeline:

1. **Retrieve & chunk** — document from feed / SEC EDGAR, split semantically.
2. **Score** — LLM (or FinBERT) assigns sentiment per chunk.
3. **Aggregate** — position-weighted into a document signal $s_i \in [-1, 1]$.
4. **Normalize** — within industry & period to remove sector trends.

UNDER THE HOOD

Map signal to a BL view (uncertainty *inverse* to signal strength):

$$q_i = \alpha \cdot s_i + \Pi_i, \qquad \Omega_{ii} = \sigma_q^2 \cdot (1 - |s_i|).$$

α calibrated by regressing past signals on realized excess returns. Strong signal ($|s_i| \approx 1$) $\rightarrow$ low uncertainty; neutral $\rightarrow$ prior dominates.

Beware staleness / look-ahead: use only text available *before* the rebalance close. After-hours summaries leaking into next-day trades = look-ahead bias.

Worked: Earnings-Call Views (100 S&P 500 names)

Prompt

"You are a sell-side equity analyst. Given this earnings-call excerpt, assign a return sentiment score from -1 (bearish) to $+1$ (bullish) over the next month. Output only the number."

- CFO "raising full-year guidance 5%, no demand softening" $\rightarrow$ score $\in [0.6, 0.9]$.
- CEO "cautious on macro, right-sizing cost structure" $\rightarrow$ score $\in [-0.6, -0.3]$.

Calibrated results ($T = 800$):

- Signal vs. one-month excess return correlation ≈ 0.08 (significant), implying an information ratio ≈ 0.25.
- Inserted into a BL optimizer ($\delta = 3$, $\tau = 0.05$): Sharpe **15% above** $1/N$ — *before* transaction costs.

Meskovskis & Kenyon (2024): an LLM-generated *SWOT* view from strategic filings supports longer horizons where price-based backtesting lacks power.

Constructing LLM-Based Factors

UNDER THE HOOD

Factor decomposition of returns:

$$r_{i,t} = \alpha_i + \sum_{k=1}^{K} \beta_{ik} f_{k,t} + \varepsilon_{i,t}.$$

Fama–French: MKT, SMB, HML (+ RMW, CMA in the 5-factor model).

Language factor (LF). Demean $s_{i,t}$ cross-sectionally, sort into quintiles:

$$\mathrm{LF}_t = R_t^{\text{top quintile}} - R_t^{\text{bottom quintile}}.$$

Does it earn alpha beyond the classics? Test $\hat{a} > 0$ in

$$\mathrm{LF}_t = a + \beta_{\mathrm{MKT}}\mathrm{MKT}_t + \beta_{\mathrm{SMB}}\mathrm{SMB}_t + \beta_{\mathrm{HML}}\mathrm{HML}_t + \beta_{\mathrm{RMW}}\mathrm{RMW}_t + \beta_{\mathrm{CMA}}\mathrm{CMA}_t + \varepsilon_t.$$

Earnings-call alpha exists *short-run* ($\leq$ 5 days) then decays as the market reads the same text.

The Factor Zoo — and Automated Factor Discovery

Multiple-testing inflation (Harvey, Liu & Zhu, 2016)

With hundreds of candidate factors tested on one dataset, $t > 2$ produces *many* false discoveries. A factor surviving **Bonferroni** or a multiple-comparison Sharpe test is far stronger evidence than raw $p < 0.05$.

LLMs can automate the search itself:

- *AlphaQuant* (Yuksel, 2025): an LLM-evolutionary loop that proposes, evaluates, and refines alpha features — replacing manual feature engineering.
- This amplifies the multiple-testing problem: the backtesting discipline that follows becomes **non-negotiable**.

Where in the Trading Stack Do LLMs Belong?

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Algorithmic trading spans HFT (microseconds, microstructure) to systematic macro (weeks–months, fundamentals). LLMs matter in the **slow** regime: where signal *quality*, not delivery *speed*, decides performance. Latency in hours, not ms.

A signal s_t drives a position $w_t = f(s_t)$ that must:

- scale with signal strength,
- be bounded (no runaway concentration),
- decay as the signal ages.

Sentiment-Driven Signal Generators

UNDER THE HOOD

Z-score the raw LLM output over a rolling window (e.g. 252 days), then size:

$$z_{i,t} = \frac{\tilde{s}_{i,t} - \hat{\mu}_s}{\hat{\sigma}_s}, \qquad w_{i,t} = \frac{z_{i,t}}{\sum_j |z_{j,t}|} \cdot W_{\max}.$$

Balanced $\pm$ signals $\Rightarrow$ dollar-neutral long-short (insulated from the market).

Sentiment decays — model it explicitly (exponential / EWMA):

$$s_{i,t} = \sum_{\tau \le t} \tilde{s}_{i,\tau}\, e^{-\lambda(t-\tau)}, \quad \lambda \approx 0.1\text{–}0.5 \text{ day}^{-1} \text{ (half-life 1–7 days).}$$

Kirtac & Germano (2024): GPT-3-class news scores predict next-day returns with above-chance directional accuracy and a high in-sample post-cost Sharpe — but single-sample Sharpes rarely replicate out of sample. A *magnitude* baseline, not a promise.

LLM-Guided RL for Order Execution

Almgren–Chriss optimal execution (linear impact):

$$C(\mathbf{x}) = \frac{1}{2}\gamma X^2 + \eta \sum_k \tau_k v_k^2 + \lambda\sigma^2 \sum_k \tau_k x_k^2.$$

permanent impact + temporary slippage + shortfall-variance penalty; front-load when risk-averse, spread uniformly when not.

RL extends this (FinRL): state $s_t =$ (inventory, time left, spread, momentum); action $a_t \in [0, X_t]$; reward $r_t = -\mathrm{cost}(a_t)$.

UNDER THE HOOD

LLM role = **state augmentation**, not replacement:

$$\tilde{s}_t = [\, s_t^{\text{micro}},\ \text{LLM-alert}_t \,], \qquad \text{LLM-alert}_t \in \{-1, 0, +1\}.$$

A negative-surprise alert triggers *accelerated* execution, cutting adverse selection.

HARLF (Coriat & Benhamou, 2025): three-tier hierarchical RL injecting LLM sentiment into the *reward* — strong risk-adjusted returns on 2018–2024 data.

Why LLMs Are *Not* Stock Pickers

Statistically significant $\neq$ economically profitable

The market-efficiency literature has drawn this line for decades.

- **Semi-strong EMH** (Fama, 1970): public info is already in prices. Large caps adjust to news in *minutes* — faster than an LLM API can read it.
- **Grossman–Stiglitz**: *some* return must accrue to informed trading, but if everyone subscribes to the same API the signal *crowds* and alpha compresses.
- **Calibration problem**: a "70% positive" LLM probability is *not* a claim the stock rises 70% of the time — requires OOS calibration to returns.

The right niche

Heterogeneous, low-coverage, novel sources — small-cap filings, emerging-market news, niche disclosures — where analyst coverage is thin and price discovery is slow. There the Grossman–Stiglitz return survives.

Backtesting: Necessary, Not Sufficient

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A backtest asks: would this strategy have made money historically? A *realistic* one accounts for frictions, data limits, and the multiple-comparison problem. Single-period in-sample Sharpe is nearly useless — the model has seen the future.

Walk-forward validation = time-series cross-validation. For fold m with step Δ:

- *Train*: $[m\Delta,\ m\Delta + T_{\text{train}})$
- *Validate*: next T_{val} (hyperparameters only)
- *Test*: next Δ — report performance **only** on concatenated test windows.

For LLM signals the "parameters" include α, decay λ, scale $W_{\max}$ — *never* tuned on the test period.

Out-of-Sample Sharpe and Transaction Costs

Primary metric — annualised OOS Sharpe:

$$\mathrm{SR}^{\mathrm{OOS}} = \frac{\bar{r}_p^{\mathrm{OOS}}}{\hat{\sigma}_p^{\mathrm{OOS}}} \cdot \sqrt{252}.$$

Rule of thumb: SR > 1.5 over ≥ 3 years OOS before paper trading.

UNDER THE HOOD

Turnover cost model, $\Delta w_{i,t} = w_{i,t} - w_{i,t-1}$: $\mathrm{TC}_t = c \sum_i |\Delta w_{i,t}|$. Daily strategy, 15% one-way turnover, $c = 0.05\%$:

$$0.15 \times 252 \times 0.05\% \approx \mathbf{1.89\%} \text{ p.a.} \Rightarrow \text{ erases marginal alpha.}$$

Cost components: bid–ask (single-digit bps large-cap, tens small-cap), market impact ($\eta x^2 + \gamma x$), financing ($> 5\%$ p.a. hard-to-borrow).
Capacity: trade $\leq$ 5–10% of ADV; earnings-call signals crowd in the first hours.

Look-Ahead and Survivorship: The Killers

Look-ahead bias — subtle LLM-specific forms:

1. **Embedding leakage** — model fine-tuned on transcripts *post-dating* the trade.
2. **Model-selection leakage** — best LLM variant chosen by full-period performance.
3. **Timestamp errors** — calls release after close; day-t open uses unavailable info.

Survivorship bias: today's S&P 500 list excludes the distressed names removed — exactly where a risk strategy drew down. Use *point-in-time* constituents.

UNDER THE HOOD

Backtest overfitting — PBO (Bailey et al., 2014) $\to 1$ for null strategies as more combinations are tried. Minimum track record length (Lopez de Prado, 2018):

$$T^* = 1 + \left(1 - \gamma_1 \mathrm{SR}^* + \tfrac{\gamma_2 - 1}{4} (\mathrm{SR}^*)^2\right) \frac{Z_{1-\alpha}^2}{(\mathrm{SR}^*)^2}.$$

Fat-tailed LLM-sentiment returns $\Rightarrow T^*$ of *several years*.

Tail Risk from Textual Signals

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Qualitative warnings — management caution, litigation, sovereign risk — shift the *tail* of the return distribution **before** they show up in prices. LLMs read this stream and feed it into volatility and tail-loss estimates in near real time.

CVaR (Expected Shortfall) at level α, a coherent, LP-minimisable risk measure (Rockafellar & Uryasev, 2000):

$$\mathrm{CVaR}_\alpha(R) = -\mathbb{E}\left[R \mid R \leq q_\alpha(R)\right].$$

UNDER THE HOOD

Textual volatility signal v_t enters a GARCH(1,1): $\sigma_t^2 = \omega + \varphi_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \kappa\, v_{t-1}$. $\kappa > 0$: elevated risk-language forecasts higher variance — fatter tails next period. Connects to NVIX (Manela & Moreira, 2017), now at the *firm* level.

CVaR-Constrained Optimization & Live Monitoring

De-risk automatically when the text turns dark:

$$\min_{\mathbf{w}} \; \mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w} \quad \text{s.t.} \quad \mathrm{CVaR}_\alpha(\mathbf{w}^\top \mathbf{r}_t) \le \bar{C},$$

$\boldsymbol{\Sigma}_t$ time-varying (DCC-GARCH). High v_{t-1} in a sector $\Rightarrow \sigma^2_{i,t} \uparrow \Rightarrow$ tighter CVaR $\Rightarrow$ automatic de-risking.

Monitoring pipeline — three layers:

- **Ingestion** — poll EDGAR, news APIs, social at configurable frequency.
- **Classification** — LLM emits structured JSON (`risk_level`, `risk_type`, `affected_positions`).
- **Action** — route critical alerts to a risk officer or a pre-approved auto-response.

Calibration: tune to the firm's portfolio; beat *alert fatigue* ($<$ 20/day for 100 positions); match latency to use (minutes EOD, seconds intraday).

Example: An 8-K Alert That Saved Half the Loss

4:17 PM — a held name files an 8-K announcing an SEC investigation into revenue recognition. Within 30s the LLM classifies:

```
{ "risk_level": "critical", "risk_type": "legal",
  "affected_positions": ["TICKER_X"],
  "summary": "SEC formal order into revenue recognition.
              Historical precedent: 20-40% downside risk." }
```

Action layer cuts the position 2.1% → 1.0% at the next open. **Next morning the stock opens −28%.**

Promise and limit

The system acted faster than any human. But it could not foresee the exact magnitude, nor whether the probe ends in no action or a restatement. It *reduces* tail exposure — human judgment still sets the thresholds.

Lecture 10 — Five Takeaways

1. **Optimizers, not replaced.** Markowitz and Black–Litterman remain the right machinery; LLMs sharpen μ from text. Extractors, not optimizers.
2. **Factor construction is the most defensible channel.** Test LLM-derived long-short alpha vs. Fama–French; correct for multiple comparisons.
3. **Efficiency limits but doesn't eliminate alpha.** Grossman–Stiglitz $\Rightarrow$ value lives in low-coverage markets; large caps are the hardest target.
4. **Backtesting discipline is non-negotiable.** Walk-forward, real costs, point-in-time data, overfitting correction — research artefact vs. deployable strategy.
5. **Real-time risk monitoring** is high-value, low-competition: structured risk signals + GARCH augmentation + continuous surveillance, with auditable value.

Next

Practical 10: build the efficient frontier, map an LLM score to a BL view, and run a cost-aware walk-forward backtest in Python.

Appendix

Extended Derivations, Benchmarks, and Further Reading

Extra / optional material — beyond the core lecture.

Appendix: Sharpe-Ratio Geometry of the Frontier

The unconstrained tangency portfolio $\mathbf{w}^{\mathrm{SR}} \propto \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ maximises

$$\frac{\boldsymbol{\mu}^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}}.$$

- The frontier (`gen_efficient_frontier.py`, deterministic, no data/network) places the tangency and $1/N$ portfolios *almost on top of one another* — a concrete picture of the DeMiguel et al. result.
- Adding the non-negativity constraint $\mathbf{w} \geq \mathbf{0}$ removes the closed form; the mean-variance program must be solved numerically as a QP.

Appendix: Recent LLM-Portfolio Literature Map

Work	Mechanism	Result
Mantshimuli & Mwamba (2025)	Multi-LLM $\rightarrow$ LSTM $\rightarrow$ BL views	Sharpe / return $\uparrow$ vs. sing
Yuksel (2025), *alpha*	LLM evolutionary tournament	Portfolio selection, large u
Yuksel (2025), AlphaQuant	LLM-evolutionary factor search	Automated alpha-feature
Coriat & Benhamou (2025)	HARLF: 3-tier hierarchical RL	Strong risk-adj. returns 20
Meskovskis & Kenyon (2024)	SWOT views from strategic filings	Long-horizon view genera
Saha, Lyu et al. (2025)	Survey of agentic PM systems	Taxonomy + open benchr

Practitioner survey: Mann (2024) — durable value sits in the *information-extraction layer*, not the decision layer.

Appendix: News, Volatility, and the EMH Evidence

- **Tetlock (2007)**: dictionary sentiment from a newspaper column predicted excess returns over a decade ago — the *incremental* value of LLMs over dictionaries is the relevant comparison; broad-coverage signals have decayed.
- **NVIX** (Manela & Moreira, 2017): text-based disaster-risk measure from newspaper archives predicted market variance and crash risk. LLMs push this to firm-level, real-time.
- **Regulatory context**: Fed SR 11-7 requires model outputs be explainable, validated, auditable — directly binding on LLM risk signals (see the RegTech chapter).

Appendix: Signal-to-View Calibration Detail

Why $\Omega_{ii} = \sigma_q^2(1 - |s_i|)$?

- It makes view *uncertainty* shrink as signal *magnitude* grows.
- Neutral signals ($s_i \approx 0$) get high Ω_{ii}, so the equilibrium prior Π_i dominates — the optimizer ignores noise.
- α is the *economic translation* from a unitless score to a return view, estimated by regressing realized excess returns on lagged signals.

Pitfall: calibrating α on the test window is model-selection leakage — it must use only data predating the evaluation window.